

ELIA RÜHLE

Technische Universität Dresden, LASR Lab ◊ elia.ruehle@lasr.org ◊ elia.ruehle@gmail.com

Research Interests: World Models, Embodied AI, Long-Horizon Planning, Robotics, Representation Learning

EDUCATION

Technische Universität Dresden

October 2023 - December 2026

M.Sc. in Computer Science with focus on Artificial Intelligence - Current GPA: 1.1/1.0 (German scale)

Thesis: Leveraging Latent Space Structure in Predictive World Models for Long-Horizon Planning

Supervisor: Prof. Roberto Calandra, Prof. Christel Baier

Technische Universität Dresden

October 2019 - September 2023

B.Sc. in Computer Science & Mathematics with graduation in Computer Science - GPA: 1.5/1.0 (German scale)

Thesis: Learning Grasp-Stability with Multi-Finger Hands and Touch in Simulation - 1.0/1.0 (German scale)

Supervisor: Prof. Roberto Calandra, Prof. Uwe Aßmann

RESEARCH EXPERIENCE

TU Dresden - Learning, Adaptive Systems and Robotics (LASR) Lab

October 2023 - January 2026

Research Assistant

Dresden

- Investigated diffusion-based planners for curiosity-driven exploration in model-based reinforcement learning
- Developed a scalable simulation framework for omnidirectional tactile sensors, enabling large-scale parallel robotic manipulation experiments in Isaac Gym
- Designed and implemented simulation pipelines for learning-based grasping with multi-finger robotic hands
- Conducted a comprehensive literature review on tactile robotic grasping and developed a unified taxonomy for learning-based grasping approaches
- Supervised by Prof. Roberto Calandra and Johannes Busch

TU Dresden - Chair of Machine Learning for Computer Vision

March 2023 - September 2023

Research Assistant

Dresden

- Conducted research at the intersection of discrete optimization and computer vision, focusing on the development and implementation of image segmentation algorithms
- Contributed to solving the multi-separator problem by designing and evaluating optimization-based algorithms using C++ and Python
- Supervised by Prof. Björn Andres

TEACHING EXPERIENCE

TU Dresden - Faculty of Computer Science

March 2023 - September 2023

Instructor

Dresden

- Weekly tutorials (~15 participants) on foundational and advanced concepts of the Python programming language
- Prepared slides and programming assignments which were solved and demonstrated during interactive live sessions

TU Dresden - Faculty of Mathematics

October 2020 - March 2022

Teaching Assistant

Dresden

- Teaching Assistant for Linear Algebra I & II (Mathematics program) and Discrete Structures (Computer Science program)
- Conducted weekly tutorials for groups of approximately 30 students, supported the design and grading of exercises, and assisted with the preparation and evaluation of written and oral examinations

SELECTED RESEARCH PROJECTS

Join Order Selection using Cardinalities and Reinforcement Learning

Database Research Group

April 2025 - September 2025

Dresden

- Investigated reinforcement learning-based query optimization by learning join ordering policies from cardinalities and query interactions.
- Analyzed the influence of hyperparameters and training data diversity on learning stability and generalization
- Achieved a 25.37 mean reward improvement during training and demonstrated improved generalization with larger and more diverse query workloads
- Supervised by Prof. Wolfgang Lehner

Decoding Emotions: Learning Feelings from Textual Data

ScaDS.AI

October 2024 – March 2025

Dresden

- Developed a transformer-based model for multi-label emotion prediction and intensity classification using pretrained RoBERTa representations
- Investigated how different pooling strategies and classification-head architectures affect model performance and generalization
- Achieved 76.57% micro-F1 on emotion classification, outperforming the baseline by 20.94%
- Supervised by Prof. Michael Färber and Prof. Simon Razniewski

Unveiling Active Learning Strategies: Similarities, Applicability to Unlabeled Data and Influence of Initial Performance

Database Research Group

April 2023 - August 2023

Dresden

- Clustered 37 Active Learning (AL) strategies to identify similarities and support systematic strategy selection across diverse machine learning tasks
- Developed a decision tree-based recommendation framework for selecting effective AL strategies based on dataset characteristics, achieving 69.2% accuracy on unseen datasets
- Supervised by Prof. Wolfgang Lehner and Dr. Julius Gonsior

SKILLS & INTERESTS

Programming Languages	Python, L ^A T _E X, Bash, C/C++, Fortran
Tools	PyTorch, NumPy, Scikit-Learn, Issac Gym, MuJoCo, Hugging Face, Linux, Git
Languages	German(native), English(fluent), Latin
Interests	Representation Learning, World Models, Robot Learning, Embodied AI, Reinforcement Learning, Long-Horizon Planning, Robotics, Continual Learning

REFEREES

Prof. Dr. Roberto Calandra

Machine Learning for Robotics (LASR Lab)

TU Dresden, Germany

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Prof. Dr. Wolfgang Lehner

Database Research Group

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Prof. Dr. Simon Razniewski

[Knowledge-Aware Artificial Intelligence]

TU Dresden, Germany

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